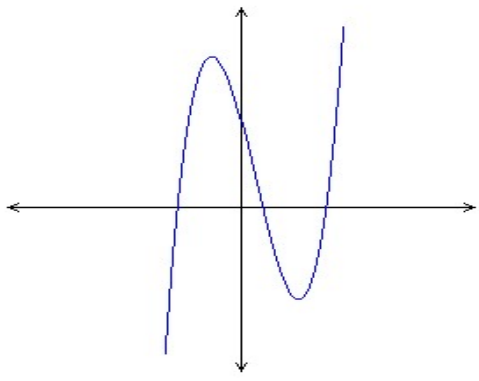
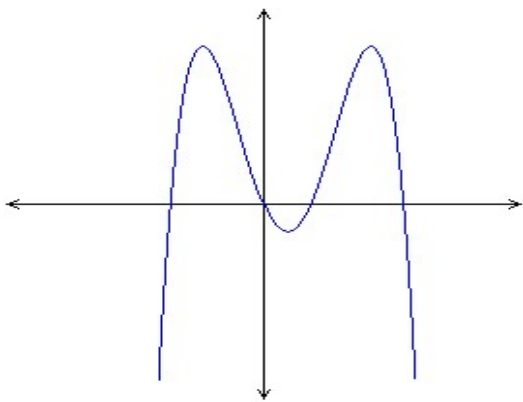
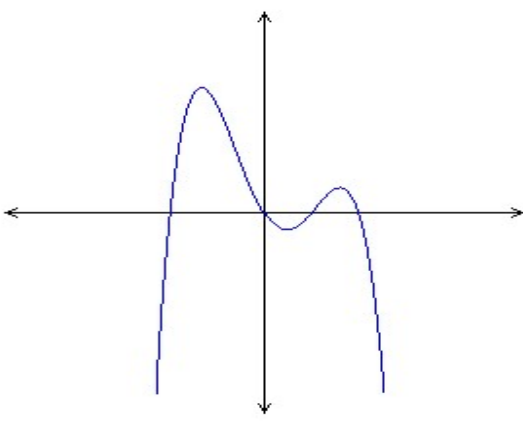
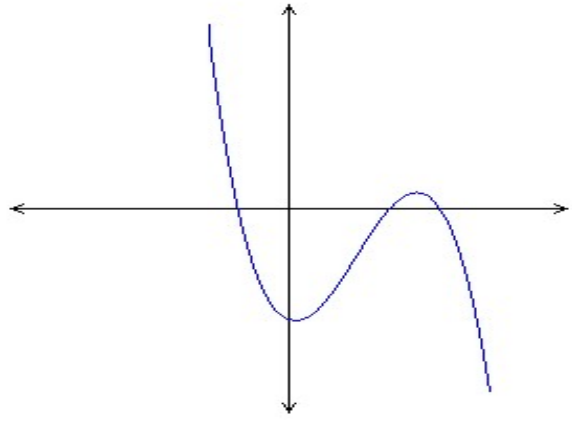


Exit Card!

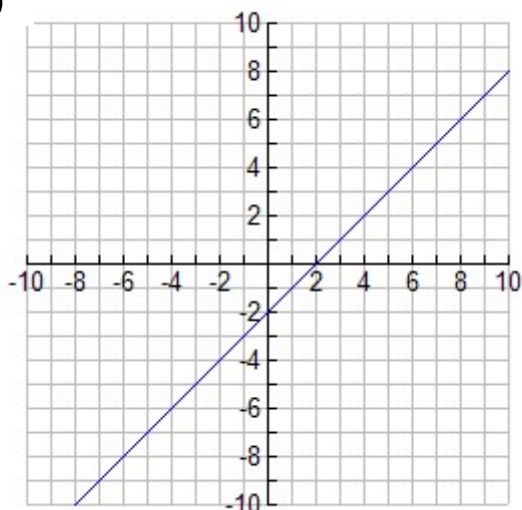
Identify number of zeros/x-intercepts/roots, the sign of the leading coefficient and describe the end behaviour. Using this information, decide if each function is cubic or quartic.

a)  # of zeros/x-int/roots: <u>3</u> Sign of Leading Coefficient: <u>(+)</u> End Behaviour: as $x \rightarrow -\infty$ $y \rightarrow -\infty$ as $x \rightarrow +\infty$ $y \rightarrow +\infty$ Type of function: <u>cubic</u>	b)  # of zeros/x-int/roots: <u>4</u> Sign of Leading Coefficient: <u>(-)</u> End Behaviour: as $x \rightarrow -\infty$ $y \rightarrow -\infty$ as $x \rightarrow +\infty$ $y \rightarrow -\infty$ Type of function: <u>quartic</u>
c)  # of zeros/x-int/roots: <u>4</u> Sign of Leading Coefficient: <u>(-)</u> End Behaviour: as $x \rightarrow -\infty$ $y \rightarrow -\infty$ as $x \rightarrow +\infty$ $y \rightarrow -\infty$ Type of function: <u>quartic</u>	d)  # of zeros/x-int/roots: <u>3</u> Sign of Leading Coefficient: <u>(-)</u> End Behaviour: as $x \rightarrow -\infty$ $y \rightarrow +\infty$ as $x \rightarrow +\infty$ $y \rightarrow -\infty$ Type of function: <u>cubic</u>

Practice:

Take a look at the following graphs and answer the questions.

a)



Domain: $\{x \in \mathbb{R}\}$

Range: $\{y \in \mathbb{R}\}$

Number of roots: 1

Roots: 2

End Behaviour:

$$\begin{aligned} \text{as } x \rightarrow -\infty & \quad y \rightarrow -\infty \\ x \rightarrow +\infty & \quad y \rightarrow +\infty \end{aligned}$$

Degree: 1

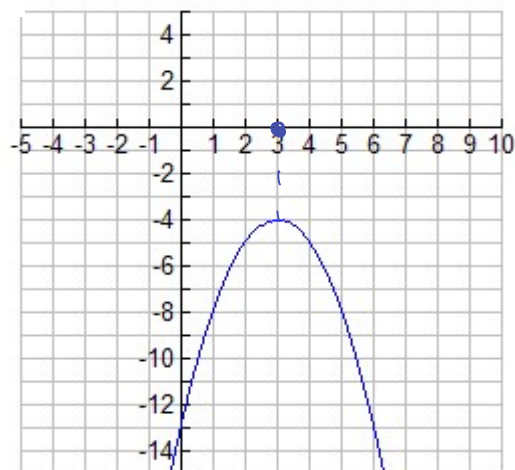
Intervals of positive: $x \in (2, +\infty)$

Intervals of negative: $x \in (-\infty, 2)$

Intervals of increasing: $x \in (-\infty, \infty)$

Intervals of decreasing: None

b)



Domain: $\{x \in \mathbb{R}\}$

Range: $\{y \in \mathbb{R} \mid y \leq -4\}$

Number of roots: none

Roots: none

End Behaviour:

$$\begin{aligned} \text{as } x \rightarrow -\infty & \quad y \rightarrow -\infty \\ x \rightarrow +\infty & \quad y \rightarrow -\infty \end{aligned}$$

Degree: 2

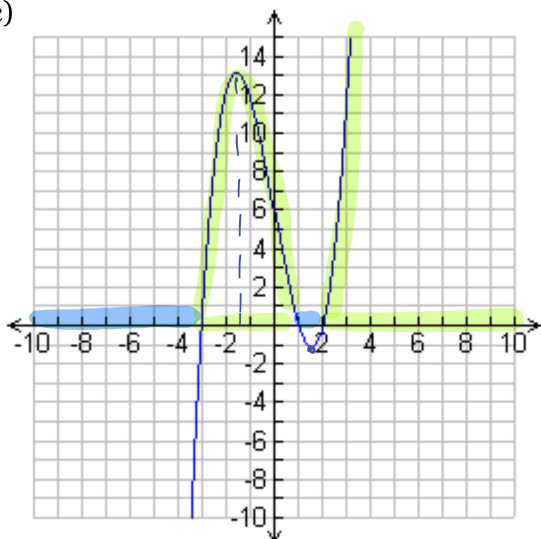
Intervals of positive: none

Intervals of negative: $x \in (-\infty, +\infty)$

Intervals of increasing: $x \in (-\infty, 3)$

Intervals of decreasing: $x \in (3, +\infty)$

c)

Domain: $\{x \in \mathbb{R}\}$ Range: $\{y \in \mathbb{R}\}$

Number of roots: 3

Roots: $-3, 1, 2$

End Behaviour:

$$\begin{aligned} \text{as } x \rightarrow -\infty \quad y &\rightarrow -\infty \\ x \rightarrow +\infty \quad y &\rightarrow +\infty \end{aligned}$$

Degree: 3 Name: Cubic

Intervals of **positive**:

$$x \in (-3, 1) \cup (2, +\infty)$$

Intervals of **negative**:

$$y \in (-\infty, -3) \cup (1, 2)$$

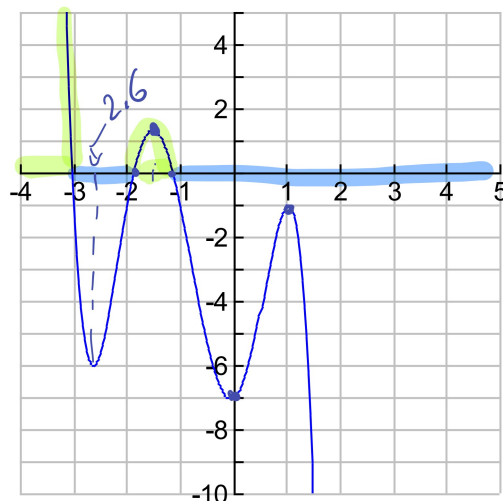
Intervals of increasing:

$$x \in (-\infty, -1.5) \cup (1.5, \infty)$$

Intervals of decreasing:

$$x \in (-1.5, 1.5)$$

d)

Domain: $\{x \in \mathbb{R}\}$ Range: $\{y \in \mathbb{R}\}$

Number of roots: 3

Roots: $-3.05, -1.8, -1.2$

End Behaviour:

$$\begin{aligned} \text{as } x \rightarrow -\infty \quad y &\rightarrow +\infty \\ x \rightarrow +\infty \quad y &\rightarrow -\infty \end{aligned}$$

Degree: Name:

Intervals of **positive**:

$$x \in (-\infty, -3.05) \cup (-1.8, -1.2)$$

Intervals of **negative**:

$$x \in (-3.05, -1.8) \cup (-1.2, +\infty)$$

Intervals of increasing:

$$x \in (-2.6, -1.5) \cup (0, 1)$$

Intervals of decreasing:

$$\begin{aligned} x \in (-\infty, -2.6) \cup (-1.5, 0) \\ \cup (1, +\infty) \end{aligned}$$